

RAPIDSEGMENT

Transparent segments. Production-ready SQL.
No black-box trees.

Strategic Segmentation & Scorecard Engine



RapidSegment

FAST. INTELLIGENT. HIGH-IMPACT SEGMENTS.

A fast, multi-segment rule discovery and
customer segmentation engine for tabular data.



FAST
& SCALABLE



HIGH IV
SEGMENTS



AUTOMATED
RULE DISCOVERY



INTERPRETABLE
& TRANSPARENT

The problem with decision trees



Hard to load

Wide tables, mixed sources,
memory spikes



Encoding tax

One-hot / label encode kills
meaning



Unreadable

Past depth 4 no one can
explain it



Same root bias

Always starts from the same
variable

Hours vs minutes

Decision Tree path

1. Load & clean
2. Encode categoricals
3. Fit tree
4. Extract leaves manually
5. Residual loops ×4
6. Translate to SQL

4–12+ hours

RapidSegment path

1. Load with UniversalDataLoader
2. Native categoricals (no encode)
3. `extract_segments()` — one call
4. SQL filters produced
5. Scorecard JSON + weights

5–15 minutes

What stakeholders actually see

Decision Tree — not human friendly

```
max_dpd_12m ≤ 15.5
├─ util_avg ≤ 0.42
│   ├─ risk_seg = Low → Leaf A
│   └─ util_avg > 0.42 → Leaf C/D
└─ max_dpd_12m > 15.5
    ├─ util ≤ 0.71 → Leaf E/F
    └─ ... 47 more nodes ...
        └─ [unreadable]
```

RapidSegment — readable SQL

SEG 1 Lift 4.8× n=2,140

```
max_dpd_12m >= 60
AND utilization >= 0.85
```

SEG 2 Lift 3.1× n=5,820

```
risk_segment = 'High'
AND max_dpd_12m >= 30
```

SEG 3 Lift 2.4× n=3,450

```
utilization >= 0.90
AND tenure_months < 12
```

How we rank features — IV & WOE (simple)

WOE — Weight of Evidence

In plain English

How much more (or less) likely an event is in this group compared with the average.

Formula

$$\text{WOE} = \ln \left(\frac{\% \text{ of non-events}}{\% \text{ of events}} \right)$$

Positive WOE → safer group Negative WOE → riskier group

IV — Information Value

In plain English

How useful is this whole feature at separating events from non-events?

Formula

$$\text{IV} = \sum (\% \text{non-events} - \% \text{events}) \times \text{WOE}$$

Higher IV = stronger predictor

Quick example — max_dpd_12m

Bin	% Events	% Non-events	WOE	Contribution
0 days	20%	55%	+1.01	Strong (safer)
30–59 days	25%	25%	0.00	Neutral
60+ days	55%	20%	–1.01	Strong (riskier)

IV ≈ 0.70 → Strong predictor (rule of thumb: IV > 0.30 is useful)

How it works — 6 steps

1

Rank & Bin

IV + Optimal Binning
on top features

2

Apriori

1-way → 2-way → 3-way
prune failures early

3

Grid Search

Sweep size × lift
pick global champion

4

Validate SQL

Translate bins to
ANSI SQL on residual

5

Residual

Remove matched rows
NULL-safe

6

Scorecard

Weights + deciles
to JSON

Four building blocks



UniversalDataLoader

CSV · Parquet · Excel
Arrow · BigQuery



BigQueryFeatureSelector

Screen IV in-cloud
Download only winners



StrategicSegmentBuilder

Finds high-lift rules
Outputs pure SQL



StrategicSegmentScore

Weights + deciles
JSON scorecard

Why teams choose it



No encoding

Categoricals stay categorical



Multi-way rules

1-, 2-, 3-way via Apriori



Zero overlap

Hierarchical residual guarantees it



ANSI SQL

Teams & production share one language



Cloud-first

Screen IV inside BigQuery



Full audit

Explain why a feature was selected/not selected at each step

From data to scorecard

```
from rapidsegment import StrategicSegmentBuilder, StrategicSegmentScore

builder = StrategicSegmentBuilder(target="default_flag", max_segments=5)

segments = builder.extract_segments(data)

scorer = StrategicSegmentScore(
    target_col="default_flag", primary_key="cust_id",
    segment_cols=[...], weight_type="ln_response"
)

model = scorer.calculate_and_export_weights(df, "model.json")
```

Stop explaining black boxes. Start shipping readable rules.

High-lift segments → pure SQL → calibrated scorecard.
Risk, credit, and marketing can all act with confidence.

github.com/B2BDA/RapidSegment · `pip install rapidsegment`



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